



C23-EC-ECII-106

23058

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FIRST YEAR EXAMINATION

ELEMENTS OF ELECTRICAL ENGINEERING

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State Faraday's first law of electromagnetic induction.
2. Define absolute permittivity and relative permittivity.
3. State Kirchhoff's current law and Kirchhoff's voltage law.
4. Write an expression for total resistance of two resistances R_1 and R_2 are connected in (a) parallel and (b) series.
5. Define impedance and power factor.
6. State voltage transformation ratio.
7. List any six specifications of transformer.
8. State the principle of DC motor.
9. List any three advantages of digital instruments over analog instruments.
10. Classify different types of batteries.

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[Contd...

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Explain charging and discharging of capacitor.
12. Explain current division rule for a two-branch parallel resistive network.
13. Explain the effect of AC flowing through pure capacitor.
14. A series R - L circuit with $R = 200\ \Omega$, $L = 0.38\ \text{H}$ is connected across 230 V, 50 Hz AC supply. Find (a) impedance, (b) current and (c) power factor and (d) voltage drop across the resistor.
15. Explain the construction and working of an auto transformer.
16. Explain the principle of operation of stepper motor.
17. Explain the working of Ramp type digital voltmeter with block diagram.
18. Explain the construction and working of lithium-ion batteries.

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